Hilbert Space Quick Summary

• **Linear vector space**: Set of vectors closed under vector addition and scalar multiplication, satisfying commutativity, associativity, distributivity, and existence of zero vector and additive inverses.

• **Hilbert space**: A complete inner-product space (completeness means every Cauchy sequence converges within the space).

• **Inner product properties**:

1. Positivity: ■ϕ | ϕ■ ≥ 0
2. Conjugate symmetry: ■ϕ | ψ■ = ■ψ | ϕ■*
3. Linearity in second argument

• **Examples**: L^2 space of square-integrable functions ψ(x) with $\int |\psi(x)|^2 \, dx < \infty$.

• **Dirac notation**: Kets |ψ■ represent state vectors, bras ■ϕ| are dual vectors; ■ϕ | ψ■ is the inner product.

• **Norm**: ||ψ|| = √■ψ | ψ■

• **Orthogonality**: ■ϕ | ψ■ = 0.

• **Orthonormality**: ■ϕ■ | ϕ■■ = δ■■.

• **Inequalities**:

Schwarz: |■ϕ | ψ■|² ≤ ■ϕ | ϕ■ ■ψ | ψ■

Triangle: ||ϕ + ψ|| ≤ ||ϕ|| + ||ψ||

• **Physical meaning**: Inner product gives probability amplitude; normalization ■ψ | ψ■ = 1 ensures total probability = 1.

Hilbert Space Notes

A Hilbert space ■ is a complete inner-product space over ■. Vectors in ■ can be added and multiplied by scalars while preserving closure. The inner product ■ϕ | ψ■ maps ■ × ■ → ■ and satisfies:
■ϕ | ψ■ = ■ψ | ϕ■* , ■ϕ | ϕ■ ≥ 0 , ■ϕ | aψ■ + bψ■■ = a ■ϕ | ψ■■ + b ■ϕ | ψ■■.
The norm ||ψ|| = √■ψ | ψ■ defines distance between vectors, enabling completeness.
In quantum mechanics, states are represented by kets |ψ■, and measurement outcomes relate to inner products. The dual vector ■ψ| acts on other kets to produce complex numbers. The condition ■ψ | ψ■ = 1 ensures proper normalization.
Common example: L^2 space of square-integrable wavefunctions ψ(x) with
■ϕ | ψ■ = ∫_{-∞}^{∞} ϕ*(x) ψ(x) dx.
Orthogonality ■ϕ | ψ■ = 0 means the states are mutually exclusive; orthonormality is ■ϕ■ | ϕ■■ = δ■■.
Important inequalities include the Schwarz and triangle inequalities, which bound norms and inner products.